

# DOPE: DOWNSTREAM OBJECTIVE-PRESERVING ENCODING FOR SYNTHETIC TABULAR DATA

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## ABSTRACT

A generator paper often reports the metric that also selected the configuration, and those native scores live on different scales. **Downstream Objective-Preserving Encoding (DOPE)** is the byte-budgeted tabular encoding and the generator that encoding writes. Retention, the shared utility column, is a null-normalized train-on-synthetic/train-on-real ratio (Esteban et al., 2017). A retention leaves the release gate closed. On 97 informative PMLB regression lineages, fit seed 11 and synthetic size  $4n$ , the CatBoost median retention is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976). The strongest classical full-panel baseline is author-default ARF. The DOPE generator’s retention minus ARF has CatBoost paired median 0.0767, with family-cluster bootstrap interval 0.0103 to 0.1957 ( $n = 97$ ; 69/0/28). Inside the density block the strongest tuned baseline is the Gaussian copula. Equivalence stays unclaimed. Forest-Flow leads on its 6-lineage block, and that Holm test stays non-significant. MFS-v2 and MFS-v3 stay null. Official tests stay sealed. The distance ledger stays outside differential privacy and outside HIPAA de-identification. The 10,240-byte cap is a release gate.

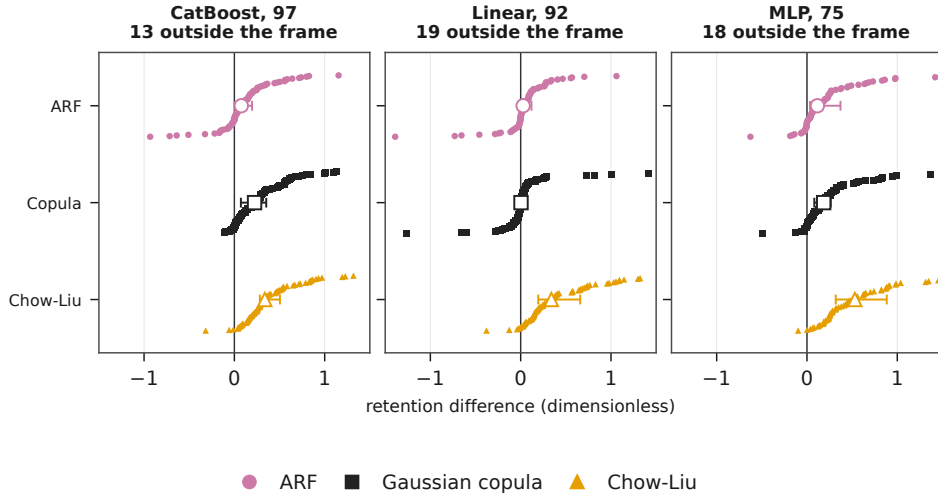


Figure 1: Paired retention differences at synthetic size  $4n$ . The axis is the DOPE generator’s retention minus the named comparator. Filled marks are lineages. The open mark is the paired median and the bar is its 95% family-cluster bootstrap interval. Circles are author-default ARF. Squares are the Gaussian copula and triangles are Chow-Liu. The count in each panel title is that auditor’s paired informative lineages. The CatBoost ARF paired median is 0.0767, family-cluster bootstrap 0.0103 to 0.1957 ( $n = 97$ ). The linear ARF interval is  $-0.0020$  to  $0.1195$  and contains zero. Equivalence stays unclaimed. The three auditors stay unpooled. Forest-Flow is a separate block and stays off this figure. No Holm test is drawn here. MFS-v2 and MFS-v3 stay null, and a retention leaves the 10,240-byte gate closed.

# 1 INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE (Xu et al., 2019), held-out log density for a Gaussian copula (Patki et al., 2016), a forest density for ARF (Watson et al., 2023), and a diffusion or CART objective for other families (Kotelnikov et al., 2023; Nowok et al., 2016). This paper separates selection from the common measurement. The supplement states the rule, the rows, and the budget for each method. CTGAN and TVAE are selected on the same validation rows and the same linear and MLP auditors that later score them. Chow–Liu and the independent marginals report no search. After those choices, the shared columns are retention and the bytes charged to the artifact. Retention is the null-normalized ratio in (2). The object under test is the encoding. The ship decision is the release gate in Section 2.4.

## 1.1 DOWNSTREAM OBJECTIVE-PRESERVING ENCODING

DOPE is the byte-budgeted encoding in Section 2 and the generator that encoding writes. Figure legends label that generator DOPE. Retention records how much of a real-trained regressor’s improvement over a null is recovered by an auditor trained on synthetic rows. That ratio is the TSTR/TRTR comparison named in Section 5. The decision those columns can support is narrow. Retention is the measurement this paper can finish. A best-generator ranking and a ship decision both stay outside that column. Master Fitness v2 (MFS-v2) and MFS-v3, defined in Section 2.4, are the release scores. Both are null on every method here. A missing component or a failed gate removes the scalar rather than publishing a small number.

## 1.2 CONTRIBUTIONS

1. **Displayed residual and sampling procedure.** The displayed profile stores a compact residual and samples a row from that artifact (Section 2, the supplement’s pipeline figure, (1)). That procedure stays off the production default and leaves the 4,168-dimensional router unused.
2. **Retention with lineage-bootstrap intervals.** The validation panel reports retention with a lineage-bootstrap interval. The algebra of a retention value is Section 2.3, (2). The supplement draws the CatBoost level of that value. Fig. 1 shows the DOPE generator’s retention minus author-default ARF, the Gaussian copula, and Chow–Liu. Charged bytes are the compactness check in Fig. 2.
3. **Paired tests inside unpooled blocks.** The density block, the neural block, and the 6-lineage Forest-Flow confirmation each keep their own pairs. Those tables and the sorted pairs are in the supplement. The linear non-separation and the Forest-Flow lead are in those blocks.
4. **A byte-budgeted release gate that stays null when a component is missing.** The release score in Section 2.4 is a geometric mean that stays null when any hard gate or positive-weight component is missing, so a retention cannot be copied into (3).

**Scope.** Official tests stay sealed. MFS-v2, MFS-v3, and PTF-v1 are null. PTF-v1 is the release KPI whose hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout. TabSyn is measured on a retained validation subset. The 100-lineage TabDDPM publication is not measured. The full Forest-Flow publication is not measured. The pre-specified privacy-attack panel is not measured. Fit-seed variance is not measured for the displayed DOPE generator. The full ablation grid is not measured. A release at the 10,240-byte tier is a separate claim from the byte cap, and that claim stays unmade here. Section 6 records what these limits do to the estimates. An unexecuted method stays outside any DOPE win.

**Nomenclature.** DOPE is the byte-budgeted encoding and the generator that encoding writes. Retention is the null-normalized ratio in (2). MFS-v2, MFS-v3, and PTF-v1 are unscored release contracts.

## 2 METHOD

### 2.1 DISPLAYED GENERATOR

The displayed profile is the generator this paper reports, and it is the DOPE artifact named in the abstract and in the figure legends. Its configuration id is `features12_steps2048`: at most 12 screened inputs, a hidden width of 16, a  $\tanh$  activation, 2,048 AdamW steps at learning rate 0.002, a refit readout, and no embedding. It is a compact neural residual. It stays outside the GAN, diffusion, and transformer families. The supplement draws that pipeline and states the training freeze and the skip refit. For a selected input vector  $x_S$ ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

There is no embedding table in (1). A separate dataset-action router uses a vector of dimension  $856 + 24 \times (128 + 10) = 4,168$ . That vector stays unused for the rows in this paper.

The supplement states the joint draw, the compile search, and the width-and-step grid. For the displayed DOPE artifacts, the charged byte count is the encoded model file plus `projection.json`. The publisher adds those two lengths. A baseline charge is the length stored for that method.

### 2.2 A FIXED ENCODING OF THE COLUMNS

The displayed generator and the dataset encoding both reduce a table to a fixed set of columns. The budgets differ, and both budgets belong to the current method.

Training screens columns by absolute rank correlation with the target and keeps at most 12. Those columns are the support  $x_S$  in (1). The artifact stores their positions in the training table. That screen is what keeps the residual inside the byte cap. It is a weak dataset key. A column left unranked by the target is dropped, and a permutation of the remaining columns changes the stored positions.

The encoding an index can store is the sketch the router already consumes. `embed-dataset` min-max normalizes each training column onto the unit interval, leaves a missing entry missing, and leaves the official test unread. Each column becomes twelve content summaries: a missing rate, moments, quantiles, a unique-value share, a tail ratio, and the absolute association with the target. Columns are sorted by those summaries, variance first, then by the rest of the summary, then by the sorted observations. A row permutation leaves the vector fixed. A column permutation leaves it fixed when the retained set has no cutoff tie; a tie there can change the pairwise summaries of the retained columns. The width of that sketch is  $10 + 12 \times 6 + 64 \times 12 + 6 = 856$ . The ten-number header records the task, the log row count, the log column count, and the shape of the target. Each of the twelve content summaries is then reduced across columns to a six-number distribution: the minimum, the lower quartile, the median, the upper quartile, the maximum, and the mean. That is the  $12 \times 6$  block. The sketch then keeps the first 64 columns in the order above and stores each column’s twelve summaries, padding a shorter table with zeros. Absolute correlations among the kept columns collapse to one further six-number distribution, so the width stays independent of the square of the column count.

The supplement states how the longer router vector sits on that sketch. The sketch stays outside the release-score components.

### 2.3 RETENTION

Let  $L_{\text{null}}$ ,  $L_{\text{real}}$ , and  $L_{\text{synthetic}}$  be the auditor’s loss on held-out validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Those held-out rows come from the official training file. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

The supplement gives the rearrangement of that ratio and an arithmetic example on invented losses. Equation (2) is a null-normalized TSTR/TRTR ratio (Esteban et al., 2017).  $L_{\text{synthetic}}$  is the loss of an auditor trained on synthetic rows and scored on real holdout rows.  $L_{\text{real}}$  is the loss of the same auditor trained on real rows and scored on those holdout rows.  $L_{\text{null}}$  is the null loss on the

holdout. The encoding and the byte-budgeted release gate are the claims. The ratio is the column they are scored on. Its delta-method variance, and the reason a small denominator is omitted, is in the supplement. A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and stays strictly positive. Otherwise the lineage is omitted. An omitted lineage stays out of the median. The three auditors are CatBoost (Prokhorenkova et al., 2018), ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. Each bar is that lineage interval.

## 2.4 THE RELEASE SCORE

The decision this project is built to make is whether an artifact may ship. Retention is the measurement this paper can finish. It is the shared column after the choices stated in the supplement. A high retention means an auditor trained on synthetic rows recovered the improvement that the same auditor obtained from real fit rows. That can be true of an artifact that is too large to ship, that misses the joint distribution, that copies training rows, or that moves the features a later model relies on. The release score keeps those failures apart from a utility median.

MFS-v2 is that score. With component  $c_i \in [0, 1]$ , weight  $w_i > 0$ ,  $\epsilon = 10^{-6}$ , and  $W = \sum_i w_i$ ,

$$\text{MFS-v2} = 100 \exp \left( \frac{1}{W} \sum_i w_i \ln(\epsilon + c_i) \right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside  $[0, 1]$ . Otherwise the scalar is null. The mean is geometric. A component near zero pulls the product down. A component that is missing, or that falls outside  $[0, 1]$ , removes the score rather than publishing a small number. An empty gate map, or any failed hard gate, does the same. Privacy has soft weight zero. A membership failure, an attribute failure, or a copied row is a gate. Utility cannot offset it. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05.

The supplement states what each of the six components measures. Compactness is  $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$ . An artifact past 10,240 bytes fails the byte gate, and the scalar stays null. Crossing the cap is a failed gate, distinct from a retention of zero.

MFS-v3 is the score a later release has to pass. MFS-v2 is unchanged. Historical receipts still pin the v2 file. The v3 weights add one component to the geometric mean. Utility transfer is 0.35. Representation closeness, driver fidelity, and distribution fidelity are each 0.15. Structure fidelity is 0.10. Coverage realism and compactness are each 0.05. Privacy stays at soft weight zero. A missing component, a failed gate, a missing prerequisite, or any other weight vector makes the scalar null. Each component enters as  $\max(c_i, \epsilon)$ . With  $W = \sum_i w_i$ ,

$$\text{MFS-v3} = \text{clip}_{[0,100]} \left( 100 \exp \left( \frac{1}{W} \sum_i w_i \ln(\max(c_i, \epsilon)) \right) \right) \quad (4)$$

when the four prerequisites hold, every hard gate passed, and every positive-weight component is finite and inside  $[0, 1]$ . Otherwise the scalar is null. A profile of all ones is 100 under (4). The four prerequisites are a complete real-versus-real control, the required size cells, verified artifact-only sampling, and locked metric implementations. The Python scorer and the native Rust v3 evaluator both take  $\max(c_i, \epsilon)$  inside the logarithm, clip the result to  $[0, 100]$ , and withhold the scalar when a prerequisite or a hard gate is missing. The historical Rust path for (3) remains that v2 formula. PTF-v1, MFS-v2, and MFS-v3 stay null on every method in these tables. The cleartext scan stays unscanned where the stored column codes are internal identifiers, so the scalar stays null. The supplement records the historical placement of  $\epsilon$ , keeps the illustration out of these tables, and applies the bounded formula to invented components. A retention stays distinct from the scalar. The compile-time proxy  $0.45u + 0.25d + 0.30j$  stays outside (3) and (4).

### 3 EXPERIMENTAL SETUP

#### 3.1 PARTITIONS AND PROVENANCE

The panel is the rights-cleared PMLB regression collection (Olson et al., 2017; Romano et al., 2022). The supplement names the simulation-style lineages in this prepared panel and records the data-lock counts, the coreset, and the file-name inventory. Official tests stay sealed, and final evaluation stays unauthorized. The study design names the endpoint partition `untouched_real_test`. The numbers below use the training-derived validation cut instead. That sealed partition stays closed, so these estimates remain validation estimates.

Fit seed 11 defines the primary displayed DOPE and legacy comparison populations. Later retained panels state their own fit-seed coverage. Sizes  $n$  and  $4n$  are both stored. Table 2 reports both. The block tables in the supplement are size  $4n$ . The supplement states the selection rule for each method. CTGAN and TVAE use the validation rows and the linear and MLP auditors that also score retention. Chow–Liu and the independent marginals report no search. TabSyn’s retained subset selected no configuration. Native objectives stay unranked across methods.

The supplement records the ARF watch-log counts. The later matched ARF publication supplies scored retention; its displayed-profile comparison median is 0.840. The new retained diagnostic panel has a separate receipt and aggregation rule. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid.

#### 3.2 STATISTICS

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is applied inside a family, separately for each test, and stays inside its own family. The Wilcoxon signed-rank test targets a pseudo-median of the nonzero paired differences. The reported estimand is the sample median and its lineage-bootstrap interval. Those two estimands stay distinct. The supplement lists the CatBoost Holm sign-test values. The density family has nine tests (three comparators times three auditors) and applies to the density rows of the supplement’s density table and of Table 1. The neural family has six tests (CTGAN and TVAE times the three auditors) and stays unpooled from the density family. The Forest-Flow family has three tests (one comparator times the three auditors) on 6 lineages and stays unpooled from either family. The Kumo, Mitra, and TabICL table in the supplement is outside those families. Its Holm adjustment is one family per auditor, each of size three: that auditor against the three density comparators. A rejection stays inside its own family. In a W/T/L count, the tie is a paired difference of zero. A non-significant Wilcoxon test stays distinct from that tie, and equivalence stays unclaimed. The supplement’s paired-difference table reports two one-sided tests at the pre-declared margin. Every filled TOST cell in that table is no. Equivalence stays unclaimed. The supplement states the Friedman and Nemenyi procedure and the size-split Holm families. The critical-difference constant for four methods at  $\alpha = 0.05$  is 2.569. The primary comparison interval resamples informative lineages from the single fit at seed 11. The predeclare names six tests per size. The size- $n$  and size- $4n$  control intervals remain in progress, wave 2, and this paper prints no number for them. These are descriptive validation tests. They stay outside the release KPI in Section 1.2, and they leave a release at the 10,240-byte tier unauthorized. MFS-v2 and PTF-v1 stay null.

#### 3.3 COMPUTE

The supplement’s compute table lists fit time, peak memory, the GPU models, GPU hours, charged bytes, and the sample-timer medians for receipts that store a timer.

### 4 RESULTS

**Strongest tuned baseline.** Author-default ARF is the strongest classical full-panel baseline at size  $4n$ . The DOPE generator’s retention minus ARF has CatBoost paired median 0.0767, family-cluster bootstrap 0.0103 to 0.1957 ( $n = 97$ ; 69/0/28). The linear interval is  $-0.0020$  to  $0.1195$  and contains

Table 1: **Strongest classical baseline at  $4n$ .** The full-panel rows are author-default ARF, with the family-cluster interval. The other rows are the strongest tuned baseline inside one block, with the paired lineage-bootstrap interval. The linear ARF interval contains zero. Equivalence stays unclaimed. Holm for the ARF rows is the size- $4n$  family of six. Holm for the other rows stays inside the block: nine density tests, six neural tests, or three Forest-Flow tests. W/T/L counts a positive, zero, or negative difference. A zero is the tie in that count.

| Block      | Auditor  | Baseline           | $n$ | Difference                | W/T/L   | Holm $p$               |
|------------|----------|--------------------|-----|---------------------------|---------|------------------------|
| Full panel | CatBoost | Author-default ARF | 97  | 0.0767[0.0103, 0.1957]    | 69/0/28 | $1.31 \times 10^{-5}$  |
| Full panel | Linear   | Author-default ARF | 92  | 0.0274[-0.0020, 0.1195]   | 61/0/31 | $8.53 \times 10^{-5}$  |
| Full panel | MLP      | Author-default ARF | 75  | 0.1153[0.0340, 0.3709]    | 57/0/18 | $4.56 \times 10^{-8}$  |
| Density    | CatBoost | Gaussian copula    | 97  | 0.224 [0.149, 0.290]      | 86/0/11 | $1.35 \times 10^{-13}$ |
| Density    | Linear   | Gaussian copula    | 92  | 0.00366 [-0.00438, 0.012] | 49/0/43 | 0.373                  |
| Density    | MLP      | Gaussian copula    | 75  | 0.185 [0.128, 0.228]      | 65/0/10 | $4.96 \times 10^{-11}$ |
| Neural     | CatBoost | TVAE               | 21  | 0.236 [0.087, 0.431]      | 19/0/2  | $1.57 \times 10^{-4}$  |
| Neural     | Linear   | TVAE               | 21  | 0.263 [0.166, 0.410]      | 20/0/1  | $1.57 \times 10^{-4}$  |
| Neural     | MLP      | TVAE               | 19  | 0.284 [0.153, 0.734]      | 17/0/2  | $1.57 \times 10^{-4}$  |
| Forest     | CatBoost | Forest-Flow        | 6   | -0.027 [-0.308, 0.00804]  | 1/0/5   | 0.375                  |
| Forest     | Linear   | Forest-Flow        | 5   | 0.012 [-0.260, 0.302]     | 3/0/2   | 0.812                  |
| Forest     | MLP      | Forest-Flow        | 4   | -0.032 [-0.227, -0.010]   | 0/0/4   | 0.375                  |

zero. The MLP interval is 0.0340 to 0.3709 ( $n = 75$ ; 57/0/18). Those three intervals are the family-cluster bootstrap of the median. The legacy simultaneous median interval stays in the supplement. Table 1 places those full-panel rows beside the strongest tuned baseline inside each block. 0.840 is the displayed-profile publication median on the paired informative lineages in that publication row. The author-default ARF CatBoost size- $4n$  cell in Table 2 is the hierarchical median on that file’s own informative count, under the median-of-seeds rule. The retained-classical table averages the three sample seeds before its lineage-bootstrap median, on its own informative count, and that receipt stays separate from Table 2. 0.940 is the lineage-bootstrap median on the density-block paired set, and the DOPE CatBoost size- $4n$  cell in Table 2 is the hierarchical median on that file’s own count. 0.586 is the density-block paired median. The Chow–Liu CatBoost size- $4n$  cell in Table 2 is the hierarchical median on that file’s own count. The supplement’s threshold table counts a different informative set. The paired median difference stays distinct from the difference of the two medians. 0.224 stays distinct from 0.940 minus 0.656.

**Both synthetic sizes.** Table 2 is the pre-registered size- $n$  endpoint beside size  $4n$ , for the methods named in that table. TabSyn stays outside it.

**CatBoost retention.** The supplement draws the level of that shared utility under CatBoost. Higher is better, and the value stays outside MFS-v2. MFS-v2 stays null on every method. Forest-Flow is a separate 6-lineage block. Its CatBoost median is higher than the paired DOPE median on that block, the Holm test stays non-significant, and that block stays off this figure.

#### 4.1 DENSITY BLOCK

The supplement holds the paired density block, including the sorted pairs. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the Chow–Liu tree, and the independent marginals. The Holm-adjusted  $p$ -values are  $1.35 \times 10^{-13}$ ,  $4.85 \times 10^{-16}$ , and  $1.17 \times 10^{-16}$ . The rank-biserial correlations are 0.890, 0.978, and 0.999. The Gaussian copula is the strongest density baseline. Wins, ties, and losses against it are 86/0/11 under CatBoost and 65/0/10 under the MLP. On the 97 lineages common to all four methods, the Friedman  $p$ -value is  $2.71 \times 10^{-44}$  and the critical difference is 0.476. All six CatBoost pairs exceed that difference.

**Retention against charged bytes.** Paired differences outside  $[-1.5, 1.5]$  are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the 10,240-byte release gate. The value of this encoding is the retention it holds at a small byte charge. Fig. 2 places that retention against charged bytes. A high retention on an artifact past the dashed byte gate still leaves the release closed. Neural and Forest-Flow artifacts stay off this figure.

Table 2: **Retention at size  $n$  and size  $4n$ .** The methods in this file, each with the three auditors. TabSyn stays outside this table. The interval is the hierarchical 95% CI. Author-default ARF and native-selected ARF are separate configurations. Forest-Flow is the native-selected cohort on its short set of lineages. Paired differences stay in the supplement. No Holm family in this table. Equivalence stays unclaimed.

| Method                      | Configuration        | Auditor  | Size | $n$ | Median  | Hierarchical 95% CI |
|-----------------------------|----------------------|----------|------|-----|---------|---------------------|
| DOPE                        | features12_steps2048 | catboost | 1n   | 97  | 0.8856  | [0.8286, 0.9702]    |
| DOPE                        | features12_steps2048 | catboost | 4n   | 97  | 0.9399  | [0.8964, 0.9832]    |
| DOPE                        | features12_steps2048 | linear   | 1n   | 92  | 0.9836  | [0.9675, 0.9944]    |
| DOPE                        | features12_steps2048 | linear   | 4n   | 92  | 0.9940  | [0.9718, 0.9967]    |
| DOPE                        | features12_steps2048 | mlp      | 1n   | 75  | 0.9713  | [0.9322, 0.9898]    |
| DOPE                        | features12_steps2048 | mlp      | 4n   | 75  | 1.0611  | [1.0048, 1.2671]    |
| GaussianCopula              | native_selected      | catboost | 1n   | 99  | 0.6156  | [0.3080, 0.7886]    |
| GaussianCopula              | native_selected      | catboost | 4n   | 99  | 0.6557  | [0.3517, 0.8501]    |
| GaussianCopula              | native_selected      | linear   | 1n   | 94  | 0.9613  | [0.9150, 0.9898]    |
| GaussianCopula              | native_selected      | linear   | 4n   | 94  | 0.9872  | [0.9629, 0.9959]    |
| GaussianCopula              | native_selected      | mlp      | 1n   | 76  | 0.8223  | [0.7178, 0.8767]    |
| GaussianCopula              | native_selected      | mlp      | 4n   | 76  | 0.9282  | [0.8548, 1.0012]    |
| Chow-Liu                    | native_selected      | catboost | 1n   | 99  | 0.4777  | [0.2629, 0.5820]    |
| Chow-Liu                    | native_selected      | catboost | 4n   | 99  | 0.5823  | [0.2709, 0.6510]    |
| Chow-Liu                    | native_selected      | linear   | 1n   | 94  | 0.6361  | [0.2609, 0.7562]    |
| Chow-Liu                    | native_selected      | linear   | 4n   | 94  | 0.6592  | [0.2824, 0.7633]    |
| Chow-Liu                    | native_selected      | mlp      | 1n   | 76  | 0.4702  | [0.1653, 0.6383]    |
| Chow-Liu                    | native_selected      | mlp      | 4n   | 76  | 0.6891  | [0.4387, 0.7726]    |
| independent_marginals       | native_selected      | catboost | 1n   | 99  | -0.0444 | [-0.0689, -0.0299]  |
| independent_marginals       | native_selected      | catboost | 4n   | 99  | -0.0117 | [-0.0418, -0.0000]  |
| independent_marginals       | native_selected      | linear   | 1n   | 94  | -0.0501 | [-0.0820, -0.0149]  |
| independent_marginals       | native_selected      | linear   | 4n   | 94  | 0.0021  | [-0.0268, 0.0191]   |
| independent_marginals       | native_selected      | mlp      | 1n   | 76  | -0.1279 | [-0.3221, -0.0180]  |
| independent_marginals       | native_selected      | mlp      | 4n   | 76  | -0.0273 | [-0.0922, -0.0001]  |
| CTGAN                       | native_selected      | catboost | 1n   | 21  | -0.0909 | [-0.4527, 0.1244]   |
| CTGAN                       | native_selected      | catboost | 4n   | 21  | -0.0813 | [-0.2047, 0.1056]   |
| CTGAN                       | native_selected      | linear   | 1n   | 21  | -0.0564 | [-0.3837, 0.1182]   |
| CTGAN                       | native_selected      | linear   | 4n   | 21  | -0.0335 | [-0.4545, 0.1241]   |
| CTGAN                       | native_selected      | mlp      | 1n   | 19  | -0.0995 | [-1.0258, 0.3118]   |
| CTGAN                       | native_selected      | mlp      | 4n   | 19  | -0.1487 | [-0.6644, 0.1280]   |
| TVAE                        | native_selected      | catboost | 1n   | 21  | 0.5977  | [0.4150, 0.7501]    |
| TVAE                        | native_selected      | catboost | 4n   | 21  | 0.6690  | [0.4711, 0.8259]    |
| TVAE                        | native_selected      | linear   | 1n   | 21  | 0.7641  | [0.5516, 0.8617]    |
| TVAE                        | native_selected      | linear   | 4n   | 21  | 0.7959  | [0.5806, 0.8750]    |
| TVAE                        | native_selected      | mlp      | 1n   | 19  | 0.6765  | [0.3096, 0.8155]    |
| TVAE                        | native_selected      | mlp      | 4n   | 19  | 0.7528  | [0.4781, 0.8551]    |
| ARF                         | author_default       | catboost | 1n   | 99  | 0.7423  | [0.5478, 0.8404]    |
| ARF                         | author_default       | catboost | 4n   | 99  | 0.8303  | [0.7000, 0.8947]    |
| ARF                         | author_default       | linear   | 1n   | 94  | 0.9004  | [0.7511, 0.9683]    |
| ARF                         | author_default       | linear   | 4n   | 94  | 0.9510  | [0.8541, 0.9832]    |
| ARF                         | author_default       | mlp      | 1n   | 76  | 0.7998  | [0.6430, 0.9097]    |
| ARF                         | author_default       | mlp      | 4n   | 76  | 0.9311  | [0.8438, 1.0093]    |
| ARF                         | native_selected      | catboost | 1n   | 99  | 0.7354  | [0.5273, 0.8462]    |
| ARF                         | native_selected      | catboost | 4n   | 99  | 0.8149  | [0.6468, 0.8909]    |
| ARF                         | native_selected      | linear   | 1n   | 94  | 0.8880  | [0.6975, 0.9659]    |
| ARF                         | native_selected      | linear   | 4n   | 94  | 0.9269  | [0.8465, 0.9775]    |
| ARF                         | native_selected      | mlp      | 1n   | 76  | 0.8146  | [0.6548, 0.9183]    |
| ARF                         | native_selected      | mlp      | 4n   | 76  | 0.9279  | [0.8370, 1.0283]    |
| ForestDiffusion/Forest-Flow | native_selected      | catboost | 1n   | 6   | 0.9694  | [0.7855, 0.9985]    |
| ForestDiffusion/Forest-Flow | native_selected      | catboost | 4n   | 6   | 1.0025  | [0.9256, 1.0362]    |
| ForestDiffusion/Forest-Flow | native_selected      | linear   | 1n   | 5   | 0.9806  | [0.7538, 1.0827]    |
| ForestDiffusion/Forest-Flow | native_selected      | linear   | 4n   | 5   | 0.9851  | [0.9772, 1.0007]    |
| ForestDiffusion/Forest-Flow | native_selected      | mlp      | 1n   | 4   | 0.9805  | [0.3176, 1.0294]    |
| ForestDiffusion/Forest-Flow | native_selected      | mlp      | 4n   | 4   | 1.0436  | [1.0035, 1.4676]    |

**Clean-license tabular transfer.** The supplement’s table is the same density block scored by Kumo large, Mitra, and TabICL. The shared map and the three sample seeds are the ones named above. These auditors stay unpooled from the CatBoost, linear, and MLP rows. The supplement shows those lineage medians and sorted paired differences. In that supplement display, squares are the Gaussian copula, triangles are Chow–Liu, and diamonds are the independent marginals.

**Release scalar on this receipt.** Kumo large averages 16 inverse-transformed quantile forecasts. The receipt scores 0 of 388 method–lineage cells (97 lineages  $\times$  4 methods), so the scalar is null. Cleartext is unscanned on 388 method–lineage cells.

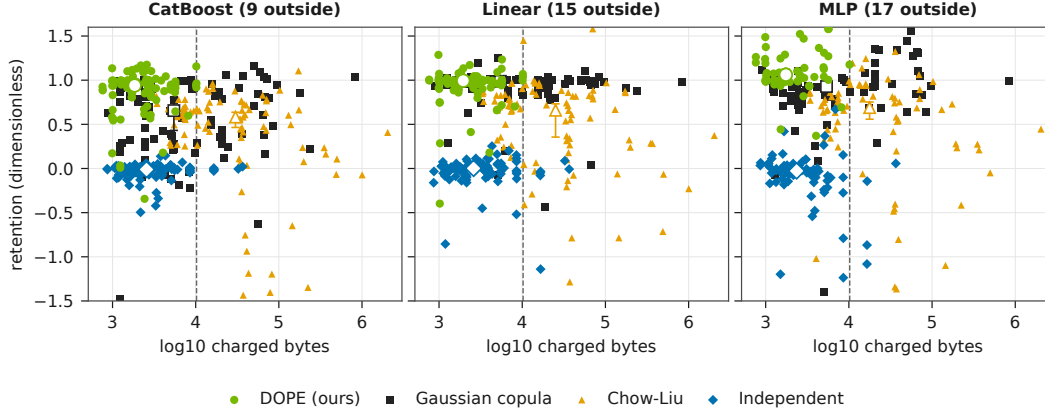


Figure 2: **Density-block retention at  $4n$  against charged bytes.** The denominator is the density-block lineages drawn in the panel. The dashed line is the release byte gate. No Holm family.

|            |                                                     |
|------------|-----------------------------------------------------|
| Kumo large | <code>row.project.output.mean.labeled.rows</code>   |
| Mitra      | <code>final.layer.norm.input.mean.self.query</code> |
| TabICL     | <code>tficl.output.mean.self.query</code>           |

**Linear non-separation.** The linear auditor leaves DOPE unseparated from the Gaussian copula (49/0/43; Holm 0.373; difference 0.00366, interval  $-0.00438$  to  $0.012$ ; rank-biserial 0.107). The interval contains zero. That non-separation leaves equivalence unclaimed. Ranks are 1.533 and 1.598 against critical difference 0.489. Chow-Liu and the independent marginals remain separated under the linear auditor and the MLP. A CatBoost separation stays local to CatBoost.

**Raw auditor loss.** Where an MLP retention sits above 1, the synthetic RMSE 0.083 is below the real RMSE 0.098. The ratio exceeds one when the synthetic-trained loss is smaller than the real-trained loss. The supplement holds the median raw loss on the same informative lineages.

**Charged bytes.** The supplement counts artifacts at or under the cap and names the two displayed DOPE artifacts over the cap.

**Fidelity and empirical privacy.** The supplement reports the density-cell classifier two-sample AUC, the marginal Kolmogorov–Smirnov distance, the pairwise correlation fidelity, alpha-precision, beta-recall, the unbalanced DCR share, and a DOMIAS AUC. The review privacy panel remains in progress, wave 2. This paper prints no number for it. Empirical resemblance stays outside differential privacy and outside HIPAA de-identification.

## 4.2 NEURAL BLOCK

**Neural block.** The supplement’s neural table stays unpooled from its density table. On the 21 CatBoost lineages, the DOPE generator’s retention median is 0.953, CTGAN’s is  $-0.081$ , and TVAE’s is 0.669. TVAE is the stronger neural baseline. Wins, ties, and losses against TVAE are 19/0/2, 20/0/1, and 17/0/2. The median favors DOPE, and DOPE still loses lineages. Every successful CTGAN and TVAE artifact in this byte count is over the cap (0/22 and 0/22). Runs that stopped at the recorded time limit are omitted. They stay outside the failure count of a finished fit. The matched TabSyn cohort remains in progress, wave 2. This paper prints no number for it. The retained TabSyn subset stays a separate cohort.

## 4.3 FOREST-FLOW

**Forest-Flow confirmation.** The supplement’s Forest-Flow table is a 6-lineage confirmation, kept separate from the density and neural blocks. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Wins, ties, and losses are 1/0/5 under CatBoost, 3/0/2 under the linear auditor, and 0/0/4 under the MLP. DOPE is lower on every paired MLP lineage. The Holm tests stay non-significant at this count, so a certified win stays unclaimed for either method. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes

(median 325,848,913.5) against a DOPE median of 2,067 on the same lineages. MFS-v2 on this block is null.

#### 4.4 RETAINED VALIDATION EVIDENCE

The supplement records the retained classical rows, the matched TabSyn subset, the legacy DOPE–ARF intervals, and the coverage table. Those cohorts stay distinct from the density block. Complete-population TabDDPM and Forest-Flow receipts stay omitted.

#### 4.5 RELEASE SCORE

**Release scalars stay null.** (3) remains the historical formula. The supplement shows invented rows of (4). The evaluation locks for a final test stay unfrozen.

### 5 RELATED WORK

The supplement organizes generators by family and credits the evaluation literature. Retention in (2) is a null-normalized TSTR/TRTR ratio (Esteban et al., 2017). The claim remains the byte-budgeted encoding and the release gate.

### 6 LIMITATIONS

**Five measurement confounds.** (i) Synthetic size  $4n$  gives the auditor more rows than the  $n$ -row real fit, which moves retention toward the generator when extra rows help. Table 2 reports both sizes. The real-row resample at size  $4n$  is the row-count control. That comparison remains in progress, wave 2, and this paper prints no number for it. (ii) A predictor-only control sits beside the synthetic rows. That comparison remains in progress, wave 2, and this paper prints no number for it. Equivalence stays unclaimed. (iii) CTGAN, TVAE, and the displayed profile’s compile search select and score on the same validation rows. (iv) Related simulation lineages can shift an interval that treats lineages as independent. (v) The displayed generator uses fit seed 11 alone, so the reported interval omits fitting uncertainty. Extra fit seeds, TabDDPM v5, the sealed official test, the stress suite, and the control, holdout-privacy, and TabSyn cell-level reductions remain in progress, wave 2. This paper prints no number for them. The supplement records the validation-cut deviation.

### 7 CONCLUSION

The claim is the byte-budgeted encoding and the release gate. Retention is the null-normalized TSTR/TRTR column those objects are scored on. On 97 regression lineages and fit seed 11, generator `features12_steps2048` has CatBoost retention 0.940 (interval 0.914 to 0.976). The 6-lineage Forest-Flow block leads on CatBoost (1/0/5) and the MLP (0/0/4). Those counts stay short of a certified win. MFS-v2 and MFS-v3 stay null. Official tests stay sealed. The 10,240-byte cap remains a release gate: 98/100 displayed artifacts are inside it, and a retention leaves the gate closed.

#### AVAILABILITY

Jankurai is the repository’s conformance auditor. A passing audit means the build, tests, and boundary proofs cleared the release floor with no high finding. It is not a utility score. It stays outside any MFS value and outside any privacy certificate.

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